

CS 255 Business Requirements Document

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- To train students for the driving test at their local department of motor vehicles.
- To provide online classes, practice tests, and on-the-road training.
- The client is Liam, the owner of DriverPass.

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- So many people fail their driving tests at the DMV.
- To provide online classes, practice tests, and on-the-road training.
- To access my data from anywhere, online as well as offline.
- Only modify or update the data if you're online.
- The system needs to run off the web, preferably over the cloud. We do not want to deal with backup and security.

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

- Have different employees at the company with different rights and roles.
- Needs to be able to download reports and some information that I can work on at home, using Excel.
- Know who made a reservation, who canceled it, who modified it last. Be able to print an activity report and figure out who is responsible.
- Our customers need to be able to make reservations for driving lessons, as well as modifying them. They should be able to make this reservation online using their account. Or they could call or visit our office to schedule an appointment with our secretary.
- We must be able to track which user is matched up with a certain driver, time, and car.

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) do this system need to run in? How fast should the system run? How often should the system be updated?

- The DriverPass system needs to run in a web browser as well as in a mobile application.

- The system needs to be loaded quickly because we are targeting “students for the driving test” (Liam) which puts the age roughly around 16-18 years old.
- The system should have regular maintenance updates around once a month to address any minor issues which may arise through regular use as well as small interface improvements. (Software Development 2024) Major updates should replace maintenance updates as needed.

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- The mobile application needs to be accessible from both the Apple Store and Google Play Store.
- The system needs to run in Chrome and Safari, as they account for 83% of the global browser market. (Conte 2025)
- The back end requires a database to maintain an appointment history, customer database, and management reports.
- Management reports need to be compatible with Microsoft Excel.

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- Usernames for different users will be unique and not case sensitive.
- Passwords for different user accounts will be case sensitive.
- You can't modify or update the data unless you're online

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- IT officer role needs to have full access over all accounts
- Ability to setup different employee rights and roles for their accounts.

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a “brute force” hacking attempt? What happens if the user forgets their password?

- The password must be 8 characters long, include one lowercase letter, one uppercase letter, and one special symbol.
- After two wrong attempts to log in, the system should prompt users to change their password through their on-file, company email account.
- After three wrong attempts to log in, lock the account and require IT administrators to unlock.

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."

- The system shall allow customers to schedule on the road time.
- The system shall allow customers to sign up for online classes.
- The system shall provide online practice tests.
- The system shall provide access to online classes with video and discussion boards.
- The system shall track who each customer is scheduled to drive with.
- The system shall provide backend reporting for management.

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- Customer account interface for tracking and utilizing different aforementioned services provided by the company.
- Administration interface for assisting customers with looking up or accessing various information within their accounts, should they call into the company.
- IT interface for changing passwords, unlocking accounts, and creating/changing/deleting company profiles.
- Management interface for overseeing the company, and access to download various reports.

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- The average user is computer literate, as the estimated age of customers is to be around 18 years old.
- The person learning to drive will be the person utilizing the services provided by this company.
- Design does not address overloaded servers from too much online traffic.

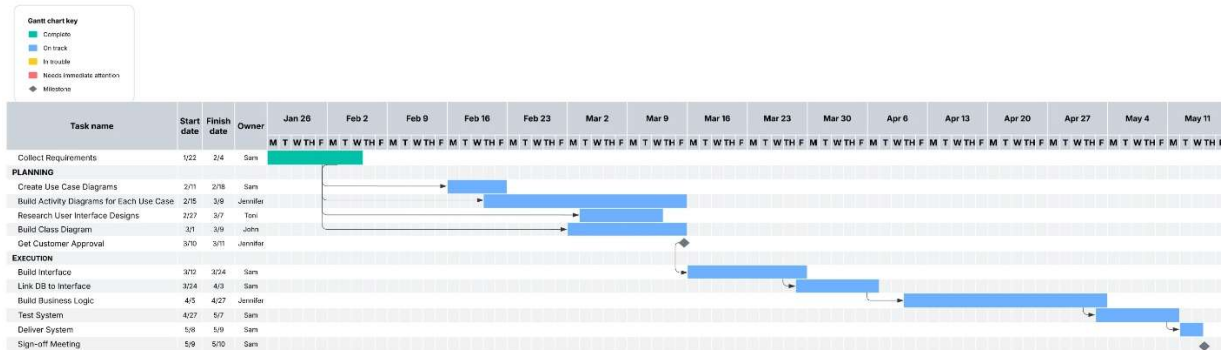
Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- This is a new company, so the budget is probably going to be very important and rather limited.
- Overhead may be limited as the company gets off the ground, so customer interface needs to be simple to use to encourage self-help.
- Smaller companies usually have older technology for their overhead staff to cut costs.

Gantt Chart

Please include a screenshot of the GANTT chart that you created with Lucidchart. Be sure to check that it meets the plan described by the characters in the interview.



References

Conte, N. (2025, May 17). *Ranked: The Most Popular Web Browsers*. Visual Capitalist.

<https://www.visualcapitalist.com/ranked-the-most-popular-web-browsers/#:~:text=Meanwhile%2C%20Microsoft%20Edge%2C%20which%20replaced,lowest%20in%20over%20a%20decade.>

Software Development. (2024, November 19). How often should you update your app? Zudu.

<https://zudu.co.uk/blog/how-often-should-you-update-your-app/#:~:text=Our%20recommendation,and%20re%20engage%20existing%20ones.&text=If%20a%20critical%20issue%20or,their%20icon%20for%20season%20events.>